Q1.

The four vertices of a parallelogram $ABCD$ have coordinates

$$A(1, 0, 2),\ B(3, -1, 5),\ C(7, 2, 4) \text{ and } D(5, 3, 1)$$

(a) (i) Find $\overrightarrow{AB} \times \overrightarrow{AD}$.

(3)

(ii) Show that the area of the parallelogram is $p\sqrt{10}$, where p is an integer to be found.

(2)

(b) The diagonals AC and BD of the parallelogram meet at the point M. The line L passes through M and is perpendicular to the plane $ABCD$.

Find an equation for the line L, giving your answer in the form $(\mathbf{r} - \mathbf{u}) \times \mathbf{v} = \mathbf{0}$.

(4)

(c) The plane Π is parallel to the plane $ABCD$ and passes through the point $Q(6, 5, 17)$.

Find the coordinates of the point of intersection of the line L with the plane Π.

(6)

(Total 15 marks)

Q2.

The plane Π has equation $\mathbf{r} \cdot \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix} = 11$ and the point Q has coordinates (1, 1, –1).

(a) Show that Q is in Π.

(1)

(b) Write down cartesian equations for the line l which passes through Q and is perpendicular to Π.

(2)

(c) The points M and N are on l, and each is 50 units from Π.

Find the coordinates of M and N.

(3)

(d) Given that the point P (5, 1, –4) is in Π, determine the area of triangle PMN.

(3)

(Total 9 marks)

Q3.

The position vectors of the points P, Q and R are, respectively,

$$\mathbf{p} = \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}$$

Determine the area of triangle PQR.

(Total 4 marks)

Q4.

The points P, Q and R have position vectors **p**, **q** and **r** respectively relative to an origin O, where

$$\mathbf{p} = \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} -3 \\ 4 \\ 20 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} 9 \\ 2 \\ 4 \end{bmatrix}$$

(a) Determine **p × q**.

(2)

(b) Find the area of triangle OPQ.

(3)

(Total 5 marks)

Q5.

Three points, A, B and C, have position vectors

$$\mathbf{a} = \begin{bmatrix} 1 \\ 7 \\ -1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{c} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} \text{ respectively.}$$

(a) Calculate (**b** – **a**) × (**c** – **a**).

(3)

(b) Hence find, to three significant figures, the area of the triangle ABC.

(3)

(Total 6 marks)